THE GYSIN HOMOMORPHISM OF AN EFFECTIVE CARTIER DIVISOR AGREES WITH THE CLASS OF ITS VIRTUAL NORMAL SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBN (sources/stacks-chow/pinned-original-source.tex line 12174).

Allowed standard prerequisites (editorial).

- The Gysin class attached to a divisor and the virtual normal sheaf. Same chapter, Definition labelled definition-gysin-homomorphism and Lemma labelled lemma-gysin-bivariant, invoked in the statement. Fixes the two classes being compared.
- Testing identities of bivariant classes on an integral base. Same chapter, Lemma labelled lemma-bivariant-zero, cited at proof lines 12185-12187. Reduces to X integral and the identity i^*[X] = c cap [X].
- The fundamental computation for the class of a virtual normal sheaf. Same chapter, Lemma labelled lemma-gysin-fundamental, cited at proof lines 12195-12198 and 12200-12203. Evaluates the class in both cases.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 02T8: sources/stacks-chow/pinned-original-source.tex, source lines 5069–5069. Same-chapter pointer into the author environment definition at source line 5069; invoked at proof line 12194 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B79: sources/stacks-chow/pinned-original-source.tex, source lines 6149–6149. Same-chapter pointer into the author environment lemma

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at source line 6149; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Same-chapter pointer into the author environment lemma at source line 6585; invoked at proof line 12186 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBM: sources/stacks-chow/pinned-original-source.tex, source lines 12076–12076. Same-chapter pointer into the author environment lemma at source line 12076; invoked at proof line 12193 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $(\mathcal{L}, s, i : D \to X)$ be a triple as in Definition 02T8. The Gysin homomorphism i^* viewed as an element of $A^1(D \to X)$ (see Lemma 0B79) is the same as the bivariant class $c(D \to X, \mathcal{N}) \in A^1(D \to X)$ constructed using $\mathcal{N} = i^*\mathcal{L}$ viewed as a virtual normal sheaf for D in X.*

Proof. We will use the criterion of Lemma 02UC. Thus we may assume that X is an integral scheme and we have to show that $i^*[X]$ is equal to $c \cap [X]$. Let $n = \dim_\delta(X)$. As usual, there are two cases.

If $X = D$, then we see that both classes are represented by $c_1(\mathcal{N}) \cap [X]_n$. See Lemma 0FBM and Definition 02T8.

If $D \neq X$, then $D \to X$ is an effective Cartier divisor and in particular a regular closed immersion of codimension 1. Again by Lemma 0FBM we conclude $c(D \to X, \mathcal{N}) \cap [X]_n = [D]_{n-1}$. The same is true by definition for the Gysin homomorphism and we conclude once again. □